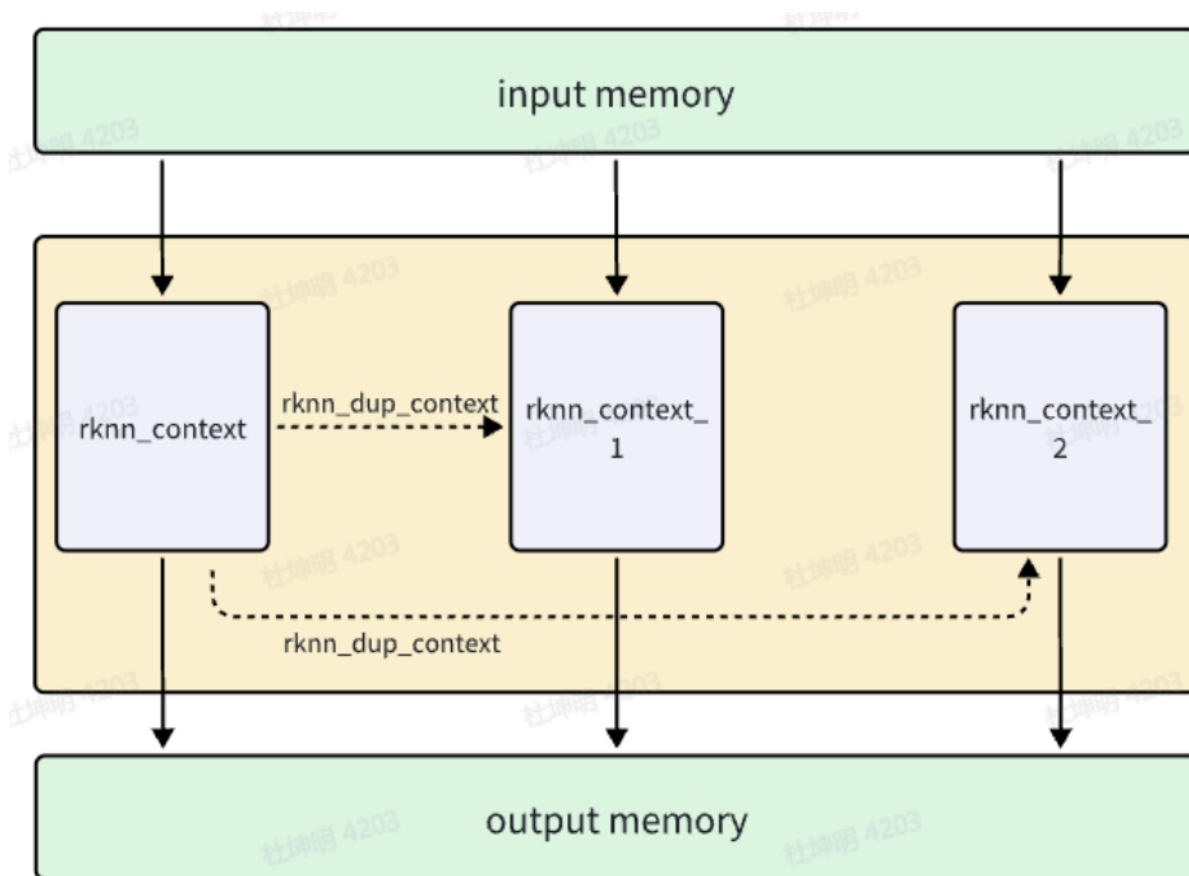


19.多Batch处理

1. 多Batch原理

RK3588 NPU内部有3个核心，RK3576 NPU内部有2个核心，为了更高效得利用多核性能，提供了多batch推理功能。当开启多batch推理时，内部会调用 `rknn_dup_context` 将 context 进行拷贝（`rknn_dup_context` 只会对 context 的Internal 进行拷贝，**Weight会复用**）。当 `rknn_batch_size=2` 时，会拷贝1份，当 `rknn_batch_size>=3` 时，会拷贝2份（同一时刻最多只有3个核心工作，为了避免内存浪费只拷贝2份）。每个 **context core_mask** 会设置成0，让多核内部自动调度。当执行 `rknn_run()` 时，内部会起一个线程池，同一时刻调用3个线程同时对3个 context 进行推理。



2. 多Batch使用方式

多batch使用方式如下：

1. Python 端开启多 batch 设置：

代码块

```
1 ret = rknn.build(do_quantization=True, dataset='./dataset.txt',  
rknn_batch_size=3)
```

为了达到最优性能，RK3588建议 rknn_batch_size 为3的倍数，RK3576建议 rknn_batch_size 为2的倍数。

2. 建议使用零拷贝接口

3. 代码测试

10_multi_batch

代码块

```
1  import numpy as np
2  import cv2
3  from rknn.api import RKNN
4
5  def show_outputs(outputs):
6      output_ = outputs[0].reshape((-1, 1000))
7      fp = open('./labels.txt', 'r')
8      labels = fp.readlines()
9      for batch, output in enumerate(output_):
10         index = sorted(range(len(output)), key=lambda k : output[k],
reverse=True)
11         top5_str = '----- Batch {}: TOP 5 -----\\n'.format(batch)
12         for i in range(5):
13             value = output[index[i]]
14             if value > 0:
15                 topi = '[{:>3d}] score:{:.6f} class:"{}"\\n'.format(index[i],
value, labels[index[i]].strip().split(':')[1])
16             else:
17                 topi = '[ -1]: 0.0\\n'
18             top5_str += topi
19         print(top5_str.strip())
20
21 def show_perfs(perfs):
22     perfs = 'perfs: {}\\n'.format(outputs)
23     print(perfs)
24
25 if __name__ == '__main__':
26
27     # Create RKNN object
28     rknn = RKNN(verbose=True)
29
30     # Pre-process config
31     print('--> Config model')
32     rknn.config(mean_values=[103.94, 116.78, 123.68], std_values=[58.82,
58.82, 58.82], quant_img_RGB2BGR=True, target_platform='rk3588')
33     print('done')
```

```

34
35     # Load model (from https://github.com/shicai/MobileNet-Caffe)
36     print('--> Loading model')
37     ret = rknn.load_caffe(model='./mobilenet_v2_deploy.prototxt',
38                           blobs='./mobilenet_v2.caffemodel')
39     if ret != 0:
40         print('Load model failed!')
41         exit(ret)
42     print('done')
43
44     # Build model
45     print('--> Building model')
46     ret = rknn.build(do_quantization=True, dataset='./dataset.txt',
47                     rknn_batch_size=3)
48     if ret != 0:
49         print('Build model failed!')
50         exit(ret)
51     print('done')
52
53     # Export rknn model
54     print('--> Export rknn model')
55     ret = rknn.export_rknn('./mobilenet_v2.rknn')
56     if ret != 0:
57         print('Export rknn model failed!')
58         exit(ret)
59     print('done')
60
61     # Set inputs
62     img_0 = cv2.imread('./dog_224x224.jpg')
63     img_0 = np.expand_dims(img_0, 0)
64     img_1 = cv2.imread('./goldfish_224x224.jpg')
65     img_1 = np.expand_dims(img_1, 0)
66     img_2 = cv2.imread('./space_shuttle_224.jpg')
67     img_2 = np.expand_dims(img_2, 0)
68
69     img = np.concatenate((img_0, img_1, img_2), axis=0) # the inputs data
70     need to be merged together using np.concatenate.
71
72     # Init runtime environment
73     print('--> Init runtime environment')
74     ret = rknn.init_runtime(target="rk3588")
75     if ret != 0:
76         print('Init runtime environment failed!')
77         exit(ret)
78     print('done')
79
80     # Inference

```

```
79     print('--> Running model')
80     outputs = rknn.inference(inputs=[img], data_format=['nhwc'])
81     show_outputs(outputs)
82     print('done')
83
84     rknn.release()
85
```

查看结果：

```
--> Export rknn model
done
--> Init runtime environment
I Target is None, use simulator!
done
--> Running Model
I GraphPreparing : 100%|██████████████████████████████████████████████████████████████████████████████| 104/104 [00:00<00:00, 19265.42it/s]
I SessionPreparing : 100%|███████████████████████████████████████████████████████████████████████████████| 104/104 [00:00<00:00, 2358.06it/s]
----- Batch 0: TOP 5 -----
[155] score:0.993652 class:"Shih-Tzu"
[154] score:0.002834 class:"Pekinese, Pekingese, Peke"
[204] score:0.002172 class:"Lhasa, Lhasa apso"
[283] score:0.000571 class:"Persian cat"
[284] score:0.000133 class:"Siamese cat, Siamese"
----- Batch 1: TOP 5 -----
[ 1] score:0.601074 class:"goldfish, Carassius auratus"
[794] score:0.062378 class:"shower curtain"
[996] score:0.062378 class:"hen-of-the-woods, hen of the woods, Polyporus frondosus, Grifola frondosa"
[927] score:0.036652 class:"trifle"
[115] score:0.028000 class:"sea slug, nudibranch"
----- Batch 2: TOP 5 -----
[812] score:0.999023 class:"space shuttle"
[148] score:0.000293 class:"killer whale, killer, orca, grampus, sea wolf, Orcinus orca"
[517] score:0.000132 class:"crane"
[833] score:0.000132 class:"submarine, pigboat, sub, U-boat"
[525] score:0.000089 class:"dam, dike, dyke"
done
(rknn-toolkit2) toneet@ubuntu:~/rknn/04_code/10_multi_batches$
```